

Daily Sequences Drill #4 — Answers & Investigations

Sheet by David Akinyele-Aje

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: *Recurrence Relations · The Characteristic Equation · Closed-Form Solutions.*

Section A Rapid Recognition

1. **Answer:** $a_2 = 5, a_3 = 14$

Method: $a_2 = 3(2) - 1 = 5. a_3 = 3(5) - 1 = 14.$

Investigate further: This is a first-order recurrence with a constant additive term — exactly the shape C1/C2 solve properly in closed form later in this sheet.

2. **Answer:** 15

Method: $a_2 = 2(1) + 1 = 3. a_3 = 2(3) + 1 = 7. a_4 = 2(7) + 1 = 15.$

Investigate further: Notice the pattern 1, 3, 7, 15, ...: each is one less than a power of 2 (2, 4, 8, 16) — worth guessing the closed form $a_n = 2^n - 1$ and checking it.

3. **Answer:** $x^2 = 5x - 6$ (i.e. $x^2 - 5x + 6 = 0$)

Method: Replace a_n, a_{n-1}, a_{n-2} with x^2, x^1, x^0 respectively in $a_n = 5a_{n-1} - 6a_{n-2}$.

Investigate further: This substitution is purely formal — it works because the recurrence is linear with constant coefficients, so a solution of the form $a_n = x^n$ must satisfy this polynomial exactly.

4. **Answer:** $x = 2, 3$

Method: $x^2 - 5x + 6 = (x - 2)(x - 3) = 0.$

Investigate further: These are the same two numbers as the recurrence's coefficients 5 (their sum) and 6 (their product) — always true by Vieta's formulas on the characteristic equation.

5. **Answer:** $a_n = A \cdot 2^n + B \cdot 3^n$

Method: Two distinct real roots r_1, r_2 give a general solution $a_n = Ar_1^n + Br_2^n$, with A, B determined by initial conditions.

Investigate further: This is the recurrence-relation analogue of a second-order linear ODE's general solution — two independent “basis” sequences, combined linearly.

6. **Answer:** A GP; $a_n = 8 \left(\frac{1}{2}\right)^{n-1}$

Method: $a_{n+1} = \frac{1}{2}a_n$ is a constant-ratio recurrence, i.e. exactly a geometric sequence with $a = 8, r = \frac{1}{2}$ (Day 3's toolkit).

Investigate further: Recognising this instantly, without reaching for a characteristic equation, is the point of B9/C6 later in this sheet.

7. **Answer:** $L = -2$

Method: $L = 3L + 4 \Rightarrow -2L = 4 \Rightarrow L = -2$.

Investigate further: L is the value at which the recurrence would stay put forever if it ever landed exactly on it — the fixed point is the anchor the whole shift-to-GP trick (B2, B3, C1, C2) is built around.

8. **Answer:** True

Method: This is exactly the general solution form derived in A5: two distinct real roots give a linear combination of two exponential sequences.

Investigate further: This stops being true the moment the roots repeat or go complex — A10 and C4 cover those two other cases.

9. **Answer:** $p = 5, q = -6$

Method: Since $a_n = 2^n + 3^n$ is built from roots 2 and 3, its characteristic equation is $(x - 2)(x - 3) = x^2 - 5x + 6 = 0$, i.e. $x^2 = 5x - 6$, giving $a_n = 5a_{n-1} - 6a_{n-2}$.

Investigate further: This is A3/A4 run in reverse — given the roots (or a sequence built from them), reconstruct the recurrence rather than solve one.

10. **Answer:** $a_n = (A + Bn)2^n$

Method: A repeated root r needs a second, genuinely independent solution alongside r^n — multiplying by n (giving nr^n) supplies it, so the general solution is $a_n = (A + Bn)r^n$.

Investigate further: This mirrors what happens with a repeated root of a polynomial more generally — an extra factor of n (or x) is needed to keep the two basis solutions linearly independent.

Section B Manipulation Drills

1. **Answer:** $A = -\frac{1}{2}, B = \frac{2}{3}$

Method: $n = 1$: $2A + 3B = 1$. $n = 2$: $4A + 9B = 4$. Doubling the first: $4A + 6B = 2$; subtracting from the second: $3B = 2 \Rightarrow B = \frac{2}{3}$. Then $2A + 2 = 1 \Rightarrow A = -\frac{1}{2}$.

Investigate further: Always check both original conditions after solving: $a_1 = -\frac{1}{2}(2) + \frac{2}{3}(3) = -1 + 2 = 1 \checkmark$, $a_2 = -\frac{1}{2}(4) + \frac{2}{3}(9) = -2 + 6 = 4 \checkmark$.

2. **Answer:** A) A GP with ratio 3.

Method: $b_{n+1} = a_{n+1} + 2 = (3a_n + 4) + 2 = 3a_n + 6 = 3(a_n + 2) = 3b_n$ — so (b_n) satisfies a pure geometric recurrence.

Investigate further: This is exactly why the shift $b_n = a_n - L$ works in general: subtracting the fixed point cancels the additive constant, leaving pure multiplication behind.

3. **Answer:** A) 241

Method: $b_1 = a_1 + 2 = 3$, and (b_n) is a GP with ratio 3: $b_n = 3 \cdot 3^{n-1} = 3^n$. So $b_5 = 3^5 = 243$, and $a_5 = b_5 - 2 = 241$.

Investigate further: Always convert back from b_n to a_n at the very end — forgetting the -2 shift-back is the most common slip in this technique.

4. **Answer:** $x^2 - x - 2 = 0$; roots $x = 2, -1$

Method: $x^2 = x + 2 \Rightarrow x^2 - x - 2 = 0 = (x - 2)(x + 1) \Rightarrow x = 2, -1$.

Investigate further: A negative characteristic root is completely normal — it produces a sign-alternating term $B(-1)^n$ in the closed form, as B5 uses directly.

5. **Answer:** A) $a_n = 2^n + (-1)^n$

Method: Check $n = 1$: $2 + (-1) = 1 \checkmark$. Check $n = 2$: $4 + 1 = 5 \checkmark$. The other options fail at $n = 1$: B) gives $2 - (-1) = 3$, C) gives $4 + 1 = 5$, D) gives $1 + 1 = 2$ — none match $a_1 = 1$.

Investigate further: With two initial conditions and two unknowns (A, B in $A \cdot 2^n + B \cdot (-1)^n$), the closed form is uniquely determined — always verify a proposed closed form against *both* given terms, not just one.

6. **Answer:** B) $a_n = 4a_{n-1} - 4a_{n-2}$

Method: Its characteristic equation is $x^2 - 4x + 4 = (x - 2)^2 = 0$ — a repeated root, discriminant $16 - 16 = 0$. The other three all have positive discriminant (A: $25 - 24 = 1$; C: $1 + 8 = 9$; D: $9 - 8 = 1$), giving distinct real roots.

Investigate further: Computing the discriminant of the characteristic equation directly ($p^2 + 4q$ for $a_n = pa_{n-1} + qa_{n-2}$) is faster than factorising every option by hand.

7. **Answer:** A) $A = 1, B = 1$

Method: $(A + B \cdot 1) \cdot 3^1 = 6 \Rightarrow A + B = 2$. $(A + B \cdot 2) \cdot 3^2 = 27 \Rightarrow A + 2B = 3$. Subtracting: $B = 1$, then $A = 1$.

Investigate further: Double-check: $a_1 = (1 + 1) \cdot 3 = 6 \checkmark$, $a_2 = (1 + 2) \cdot 9 = 27 \checkmark$.

8. **Answer:** 405

Method: $a_n = (1 + n)3^n$. $a_4 = (1 + 4) \cdot 3^4 = 5 \times 81 = 405$.

Investigate further: The repeated-root closed form grows faster than a plain exponential r^n because of the extra factor of n — compare $a_4 = 405$ here to what a plain $c \cdot 3^4 = 81c$ would need ($c = 5$) to match, which is a coincidence of this specific example, not a general shortcut.

9. **Answer:** A) The AP toolkit — it's simply $a_n = 2 + 5(n - 1)$.

Method: A constant additive step ($+5$ every time, no multiplication) is precisely Day 1's arithmetic sequence definition — the characteristic-equation machinery is unnecessary (and its "fixed point" equation $x = x + 5$ has no solution, signalling this recurrence sits outside that method's normal use case).

Investigate further: Recognising when a lighter toolkit already solves the problem is as important as knowing the heavier one — see Top 5 Pattern 4.

10. **Answer:** B) A repeated real root.

Method: $x^2 - 6x + 9 = (x - 3)^2 = 0$ — discriminant $36 - 36 = 0$.

Investigate further: This is the same equation structure as B7's recurrence, $a_n = 6a_{n-1} - 9a_{n-2}$ — worth noticing the connection.

Section C Substitution & Structure

1. **Answer:** A) $b_n = a_n - 1$

Method: Fixed point: $L = 4L - 3 \Rightarrow -3L = -3 \Rightarrow L = 1$. So $b_n = a_n - 1$, giving $b_{n+1} = a_{n+1} - 1 = (4a_n - 3) - 1 = 4a_n - 4 = 4(a_n - 1) = 4b_n$.

Investigate further: The fixed point of $a_{n+1} = pa_n + c$ is always $L = \frac{c}{1-p}$ (for $p \neq 1$) — worth memorising as a one-line formula rather than resolving $L = pL + c$ from scratch every time.

2. **Answer:** A) 1

Method: At $n = 1$: $a_1 = 1 + c \cdot 4^0 = 1 + c = 2 \Rightarrow c = 1$.

Investigate further: c here is exactly b_1 from the shift-to-GP method (C1) — the two approaches (“closed form with unknown constant” vs. “shift to a GP”) are the same technique in different notation.

3. **Answer:** A) $a_n = 3a_{n-1} + 10a_{n-2}$

Method: $p = 5 + (-2) = 3$. $q = -(5)(-2) = 10$. Check via the characteristic equation directly: $(x - 5)(x + 2) = x^2 - 3x - 10 = 0 \Rightarrow x^2 = 3x + 10 \Rightarrow a_n = 3a_{n-1} + 10a_{n-2}$.

Investigate further: This is A9's technique in general form — always rebuild the characteristic equation from the roots directly as a check, rather than trusting the $p = r_1 + r_2$, $q = -r_1r_2$ formulas from memory alone.

4. **Answer:** C) Two complex conjugate roots, $1 \pm 2i$.

Method: Discriminant $= (-2)^2 - 4(1)(5) = 4 - 20 = -16 < 0$, so the roots are complex: $x = \frac{2 \pm \sqrt{-16}}{2} = \frac{2 \pm 4i}{2} = 1 \pm 2i$.

Investigate further: A negative discriminant doesn't mean “no solution” for a recurrence the way it might for a real-world root-finding problem — the recurrence still has a well-defined (complex-valued, or real-oscillating) closed form, explored further in D3.

5. **Answer:** B) The homogeneous method plus a particular solution $C \cdot 3^n$.

Method: The constant-shift trick only cancels a *constant* forcing term — here the forcing term 3^n grows with n , so no fixed constant L can absorb it. The standard fix is to add a particular solution of the same form as the forcing term ($C \cdot 3^n$) to the homogeneous general solution.

Investigate further: This is the recurrence-relation analogue of “guess a particular solution matching the forcing function's form” from differential equations — the same idea, one level more discrete.

6. **Answer:** A) $a_{n+1} = a_n + 7$

Method: A) is a constant additive step — an AP, Day 1’s toolkit exactly. B) is a constant-ratio step — a GP (Day 3). C) needs the shift-to-GP trick (still not a plain AP). D) genuinely needs the full characteristic equation.

Investigate further: This question is the whole sheet’s “triage” skill in miniature: always ask whether Day 1 or Day 3’s lighter toolkit already applies before reaching for Day 4’s heavier machinery.

7. **Answer:** B) Linear growth in n (or constant) — effectively an AP.

Method: A repeated root $r = 1$ gives closed form $a_n = (A + Bn)(1)^n = A + Bn$ — exactly linear in n (an AP) if $B \neq 0$, or constant if $B = 0$.

Investigate further: This is a genuinely elegant full-circle result: the “two independent exponential terms” structure of a general second-order recurrence collapses back down to Day 1’s arithmetic sequence exactly when both roots equal 1 — see D4 for the reverse direction (which (p, q) forces this).

8. **Answer:** A) $q = -1$ always, $p = r + \frac{1}{r}$ can be any value with $|p| \geq 2$.

Method: $q = -r_1 r_2 = -r \cdot \frac{1}{r} = -1$, independent of r . And $p = r + \frac{1}{r}$, which by AM–GM-style reasoning satisfies $|r + \frac{1}{r}| \geq 2$ for any nonzero real r (with equality only at $r = \pm 1$).

Investigate further: The bound $|r + \frac{1}{r}| \geq 2$ is the same inequality that appears throughout the Algebra pillar’s toolkit — a nice example of one pillar’s tool resurfacing to describe a genuinely different structure (reciprocal characteristic roots) here.

Section D Challenge

1. **Answer:** $\frac{5}{3}$

Method: Characteristic equation: $2x^2 = x + 1 \Rightarrow 2x^2 - x - 1 = 0 = (2x + 1)(x - 1) \Rightarrow x = 1, -\frac{1}{2}$. General solution: $a_n = A + B(-\frac{1}{2})^n$. Using $a_1 = 1$: $A - \frac{B}{2} = 1$. Using $a_2 = 2$: $A + \frac{B}{4} = 2$. Subtracting: $\frac{3B}{4} = 1 \Rightarrow B = \frac{4}{3}$, then $A = 1 + \frac{2}{3} = \frac{5}{3}$. As $n \rightarrow \infty$, $(-\frac{1}{2})^n \rightarrow 0$, so $a_n \rightarrow A = \frac{5}{3}$.

Investigate further: Numerically: 1, 2, 1.5, 1.75, 1.625, 1.6875, ... — oscillating and visibly closing in on $1.6 = \frac{5}{3}$. A root of exactly 1 in an otherwise-decaying recurrence always signals convergence to a finite, computable limit, never to 0.

2. **Answer:** A) 405

Method: Characteristic equation $x^2 - 6x + 9 = (x - 3)^2 = 0$ — repeated root 3. Solving $a_n = (A + Bn)3^n$: $n = 1$: $(A + B)(3) = 1 \Rightarrow A + B = \frac{1}{3}$. $n = 2$: $(A + 2B)(9) = 6 \Rightarrow A + 2B = \frac{2}{3}$. Subtracting: $B = \frac{1}{3}$, so $A = 0$. Thus $a_n = \frac{n}{3} \cdot 3^n = n \cdot 3^{n-1}$. $a_5 = 5 \times 3^4 = 5 \times 81 = 405$.

Investigate further: Check the pattern directly: $a_n = n \cdot 3^{n-1}$ gives 1, 6, 27, 108, 405, ... — verify $a_3 = 6a_2 - 9a_1 = 6(6) - 9(1) = 27$ ✓ matches the recurrence.

3. **Answer:** A) The sequence oscillates in sign while its magnitude grows like 2^n .

Method: $x^2 = -4 \Rightarrow x = \pm 2i$, purely imaginary roots of magnitude 2. Testing directly with $a_1 = 1, a_2 = 0$: $a_3 = -4a_1 = -4$, $a_4 = -4a_2 = 0$, $a_5 = -4a_3 = 16$, $a_6 = -4a_4 = 0$, $a_7 = -4a_5 = -64$ —

the nonzero terms alternate sign and grow by a factor of 4 every two steps (i.e. magnitude 2 per step), exactly matching complex roots of modulus 2.

Investigate further: This is the general pattern for complex characteristic roots $re^{i\theta}$: the real sequence's magnitude grows like r^n , while it oscillates with a “period” related to θ — purely imaginary roots ($\theta = 90$) give the fastest possible sign-oscillation, a full cycle every 4 steps.

4. **Answer:** A) $(p, q) = (2, -1)$

Method: Linear-in- n behaviour needs a repeated root of exactly 1: $(x-1)^2 = x^2 - 2x + 1 = 0$, giving $p = 2, q = -1$ (matching $x^2 - px - q = 0$). Checking D: $(p, q) = (3, -2)$ gives $x^2 - 3x + 2 = (x-1)(x-2) = 0$ — roots 1, 2, distinct, so the general solution $A + B \cdot 2^n$ is linear in n only in the degenerate special case $B = 0$, not for suitable general initial conditions.

Investigate further: This is C7's result read backwards — given the desired long-term behaviour (linear growth), work out which characteristic equation, and hence which (p, q) , produces it.

5. **Answer:** A) $a_n \rightarrow 0$ always.

Method: $a_n = Ar_1^n + Br_2^n$ (or the repeated-root analogue). Since $|r_1| < 1$ and $|r_2| < 1$, both $r_1^n \rightarrow 0$ and $r_2^n \rightarrow 0$ as $n \rightarrow \infty$, regardless of the constants A, B (which are fixed by the initial conditions but don't affect whether the limit is 0).

Investigate further: This is the recurrence-relation version of Day 3's convergence condition $|r| < 1$ for a GP — both roots need to be inside the unit circle in magnitude, not just one. Tomorrow's sheet studies exactly this kind of boundedness and convergence behaviour in general.

Top 5 Patterns — Worksheet #4

- Characteristic equation: turn $a_n = pa_{n-1} + qa_{n-2}$ into $x^2 - px - q = 0$.** The entire toolkit starts here. (see A3, A4, A9, B4, C3)
- Distinct real roots $\rightarrow a_n = Ar_1^n + Br_2^n$; repeated root $\rightarrow a_n = (A + Bn)r^n$.** Always check the discriminant first to know which shape applies. (see A5, A10, B5, B7, C7)
- First-order recurrences $a_{n+1} = pa_n + c$ shift to a pure GP via the fixed point $L = \frac{c}{1-p}$.** Let $b_n = a_n - L$, solve the GP, shift back. (see A7, B2, B3, C1, C2)
- Recognise the degenerate cases first.** Constant-difference recurrences are Day 1's AP; constant-ratio recurrences are Day 3's GP — don't reach for the characteristic equation when a lighter toolkit already solves it. (see A6, B9, C6)
- Complex characteristic roots signal oscillation with exponentially changing magnitude,** not simple convergence or divergence. (see C4, D3)

Common Traps — Worksheet #4

- Forgetting to check for a repeated root before using the two-distinct-roots closed form.** A zero discriminant needs $(A + Bn)r^n$ instead. (see B6, B10)
- Mixing up the sign convention when reading p, q off from given roots.** Remember $x^2 - px - q = 0$, not $x^2 + px + q = 0$. (see C3, D4)

- 3. Applying the constant-shift trick to a recurrence with a non-constant forcing term.**
 $b_n = a_n - L$ only cancels a fixed constant — it does nothing against a term like 3^n . (see C5)
- 4. Assuming every recurrence converges to a finite limit.** Convergence needs *every* characteristic root to have magnitude strictly less than 1. (see D5)
- 5. Solving for A, B using only one initial condition.** A second-order recurrence's closed form has two free constants and needs two equations to pin them both down. (see B1, B5, B7)

“Speed comes from recognition, not from rushing. Every shortcut is a pattern you have internalised.”

Found a mistake or want to contribute a sheet?
Visit the open-source project at github.com/The-CerealDev/speed-maths